

Most companies think moving to the cloud makes them secure.. but in reality, one small misconfiguration can expose millions of users' data to the internet.

Introduction to Cloud Security

- Cloud Security = Protecting data, applications, and infrastructure in cloud environments
- Shared Responsibility Model → Cloud provider + User both responsible
- Covers:
- Data protection, Identity & Access Management (IAM), Network security, Monitoring & logging

Major Cloud Providers

- Amazon Web Services (AWS)
- Most widely used cloud platform
- Microsoft Azure Strong enterprise integration
- Google Cloud Platform (GCP)
- Known for AI/ML and data services

Shared Responsibility Model

- Cloud Provider Secures:
- Physical data centers
- Hardware & infrastructure
- User Secures:
- Data
- Access control (who can access what)
- Application configurations
- “Security of the cloud” vs “Security in the cloud”

Common Cloud Threats

- Data Breaches → Sensitive data exposed publicly
- Account Hijacking → Stolen credentials (phishing, weak passwords)
- Insider Threats → Employees misusing access
- Insecure APIs → Poorly protected backend endpoints
- DDoS Attacks → Overloading cloud services
- Malware Injection → Uploading malicious code to cloud apps

Common Cloud Misconfigurations

- Publicly exposed storage
(e.g., open S3 buckets)
- Weak IAM policies (too many permissions)
- No encryption (data stored in plain text)
- Default credentials not changed
- Misconfigured firewalls/security groups
- Lack of logging & monitoring

Identity & Access Management (IAM)

- Controls who can access resources
- Principle of Least Privilege (PoLP) → Give minimum access
- Use:
 - Strong passwords
 - Multi-Factor Authentication (MFA)
 - Role-based access

Basic Cloud Security Practices

- Enable encryption (data at rest & in transit)
- Regularly audit permissions
- Use firewalls & security groups
- Enable logging & monitoring
- Backup critical data

Why Cloud Security Matters

- Businesses rely on cloud for everything
- One mistake = huge financial + reputational loss
- Real-world breaches often caused by simple errors